

RTU Imp topic for 4th Semester

DMS :-

unit 1 Question based on inclusion and exclusion property in 4 marks and (10 marks)

Equivalence Relation

types of function

mathematical induction rule 4 marks and (10 marks)

unit 2 Propositional logic - totology , contradiction , quantifiers, question based on prove two propositional logic

CNF and DNF

Finite state machine

unit 3 POSET , hasse diagram - lower and upper bound

Lattice and it's property

bounded , complemented lattice and binomial theorem formula

recurrence relation solution for homogenous or non homogenous (10 marks)

generating function in (4 or 10 marks)

unit 4 Binary operation Alegabric properties - commutative, closure, associative, identity and inverse

group ,semi group and monoid for prove

Abelian group : prove 10 marks

Normal group : prove

ring field 10 marks

Isomorphism and homomorphism of group

unit 5 definition of different graphs

Hamiltion and eulerian graph

Isomorphism and homomorphism of graph

chromatic number (2 marks but imp)

Dijkshtra algorithm for shortest path (10 marks)

MEFA Imp topics

Unit 1 Deductive and inductive analysis, Scope and nature of economics, managerial economics, National income , GDP,

Unit 2 Demand and supply function, demand and supply elasticity, demand forecasting,

Unit 3 Cost and production function, Different types of cost, long run and short run in cost

Unit 4 Comparison between all market (oligopoly, monopoly, monopolistic, perfect)

Unit 5 Solvency , liquidity and turn over ratio, Balance sheet, short definition of all financial analysis

TC:-

unit 1 (Imp for 2 marks) Definition and Importance, Process of TC, Aspects(Competence) of TC, Linguistic Intelligence and Howard Gardner's theory, LSRW Skills, forms of TC, **Styles and Features of TC** ,

unit 2 technical Text types and Its process to read and write, **Note making and its types, guidelines for good technical writing**, document design process and information collection ,Types of Technical documents

unit 3 Grammar rules question related to tense or basic grammar(4marks), Minutes of meeting(MoM),

unit 4 Types of Articles And Process of Technical Proposal, Report and resume writing,

MPI:-

unit 1 Architecture and pin diagram of 8085, Flag registers and specific register, Multiplexing and demultiplexing, Bus system

Unit 2 Addressing mode, Instructions based on size, Timing diagram

Machine cycle and instruction cycle, time delay

Imp instructions: STA , LDA , LDX, LXI, MOV, MVI, LHLD, ADI, DAD, JMP, CALL

Unit 3 Imp programs

- a) Addition of two number 8 bit data and store at 16 bit address
- b) Compare a value with fix number
- c) Add two numbers (simple addition)
- d) Multiply of two numbers 8 bit
- e) 2's complement of BCD number
- f) Square of a number
- g

8259 interrupt controller - block and pin diagrams and specific pin description, types of interrupts

Unit 4 8279 Block and pin diagram and specific pin description , microprocessor interfacing, 8255 PPI , basic pins of 8254 and 8279

Unit 5 LCD and multiplexed display, USART 8251 Diagram , RS232C & RS422A, IEEE 488

DBMS :-

unit 1 File system v/s Database , advantages of DB, Attributes and Entity, strong v/s Weak entity, key constraints, relationship, generalization and Aggregation

Unit 2 queries based on Selection, projection , **Join - inner and outer join**, intersection, UNION, Nested and correlated queries, ODBC And JDBC , Triggers and active DB, dynamic and embedded DB

Unit 3 normal form, 3NF and BCNF, definiton of Schema Refinement, funcitonal dependency and types

Unit 4 Transaction and steps, ACID properties, definition of scheduale, serializability - conflict and view, Cascading definiton,

Unit 5 Shadow paging and log based recovery, deferred and immidiate log Deadlock handling, dirty reading, timestamp based protocol, abort , rollback and terminate, starvation problem